



CHEM F111 : General Chemistry

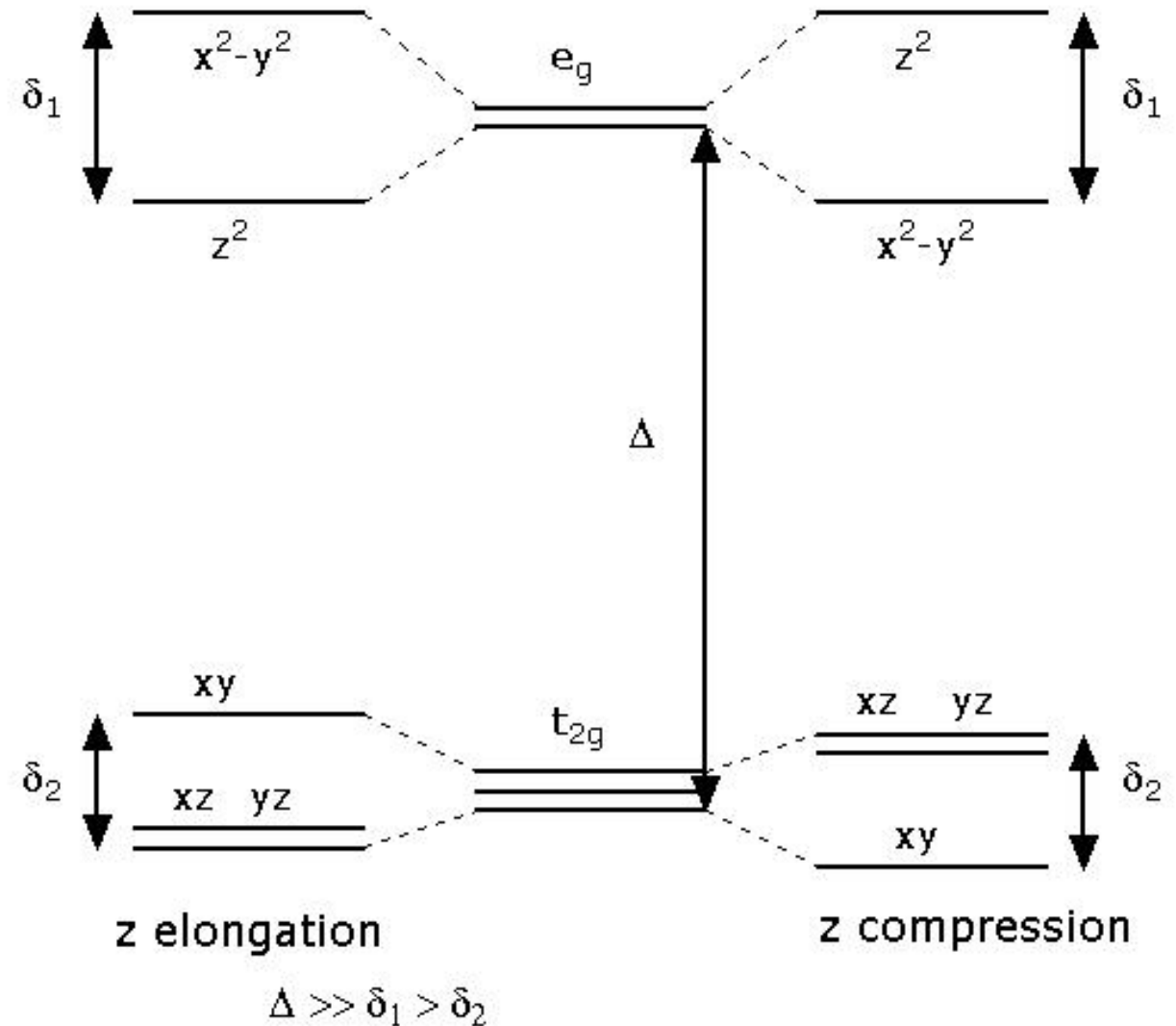
Lecture 21

Inorganic Chemistry

Discussed topics.....



- ✓ Tetragonal distortion
 - ✓ Natural distortion
 - ✓ Jahn-Teller distortion

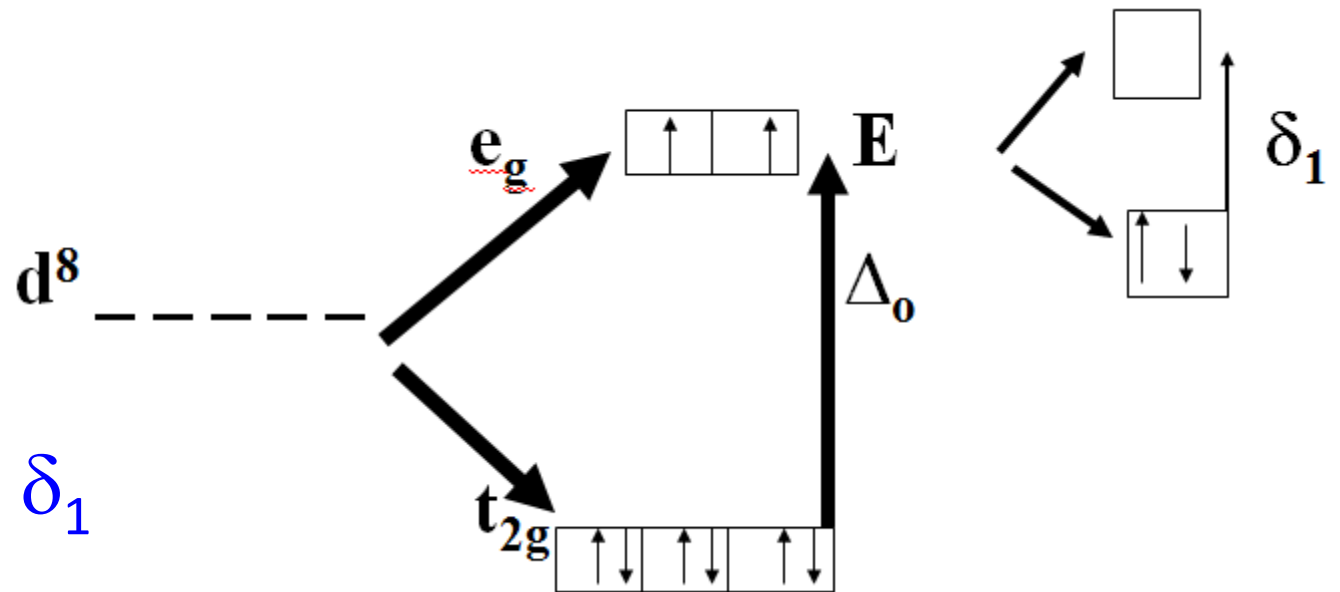


Attempts to make $[\text{Ni}(\text{CN})_6]^{4-}$ always yield $[\text{Ni}(\text{CN})_4]^{2-}$

Here Δ_o is large and hence δ_1 is large and more than the pairing energy

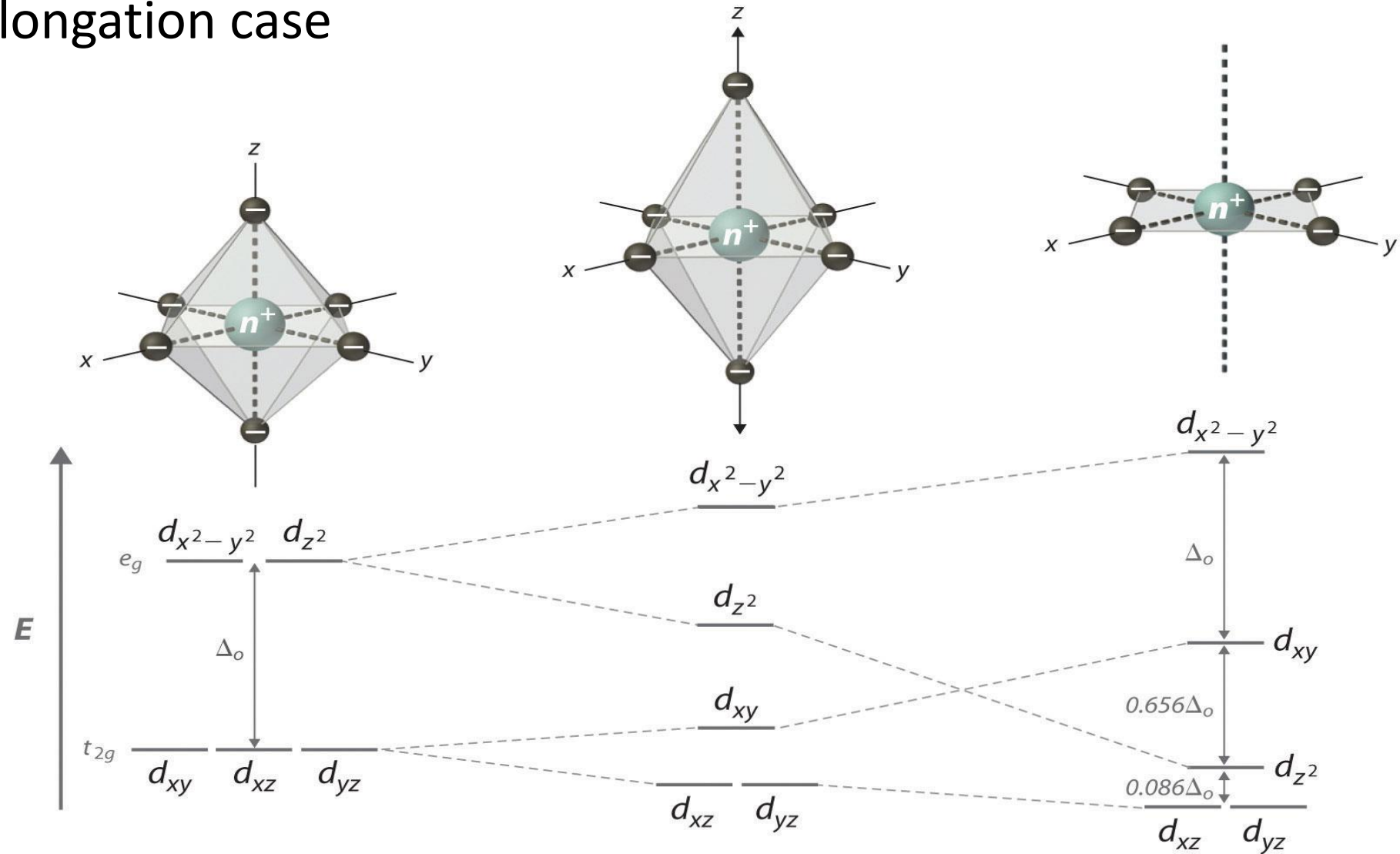
The e_g orbital which is occupied is d_z^2

This is due to pairing energy $< \delta_1$

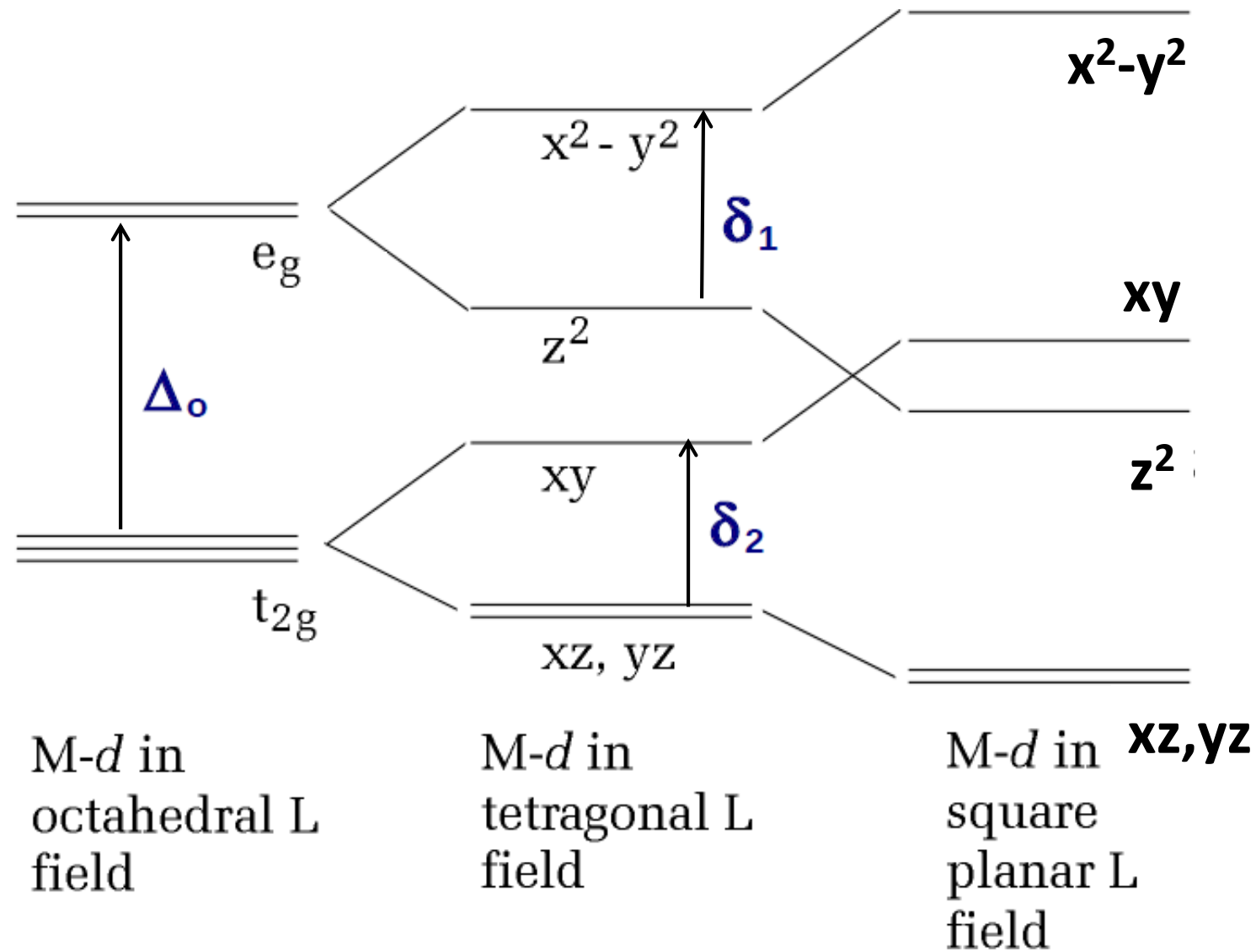


Extreme elongation

In an extreme elongation case



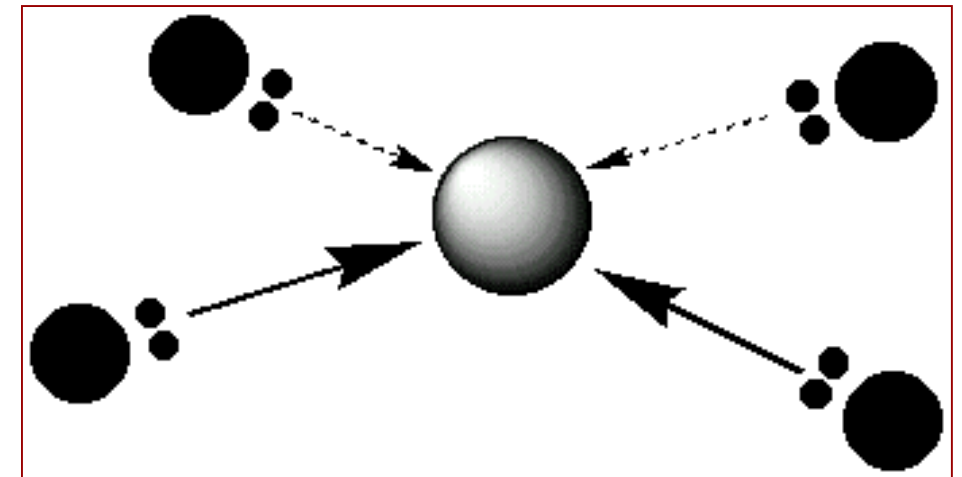
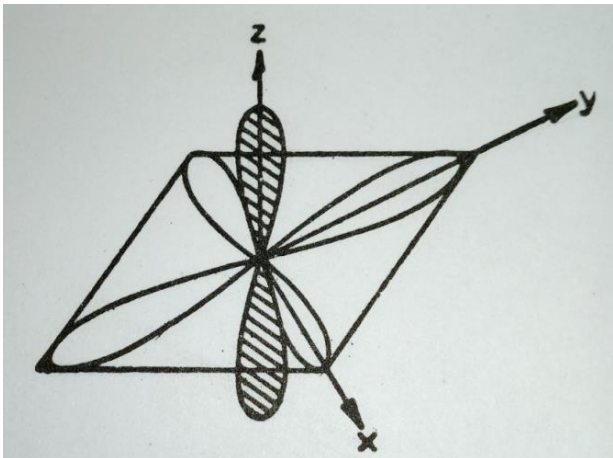
Square planar energy level



d^8 square planar system



- 1) In this case there is no unpaired electron; diamagnetic.
- 2) d_{z^2} orbital is fully occupied whereas $d_{x^2-y^2}$ orbital is completely free.
- 3) Ligands approach only through the x,y directions and z direction is free of ligands due to high repulsion.
- 4) This gives rise to Square planar configurations.

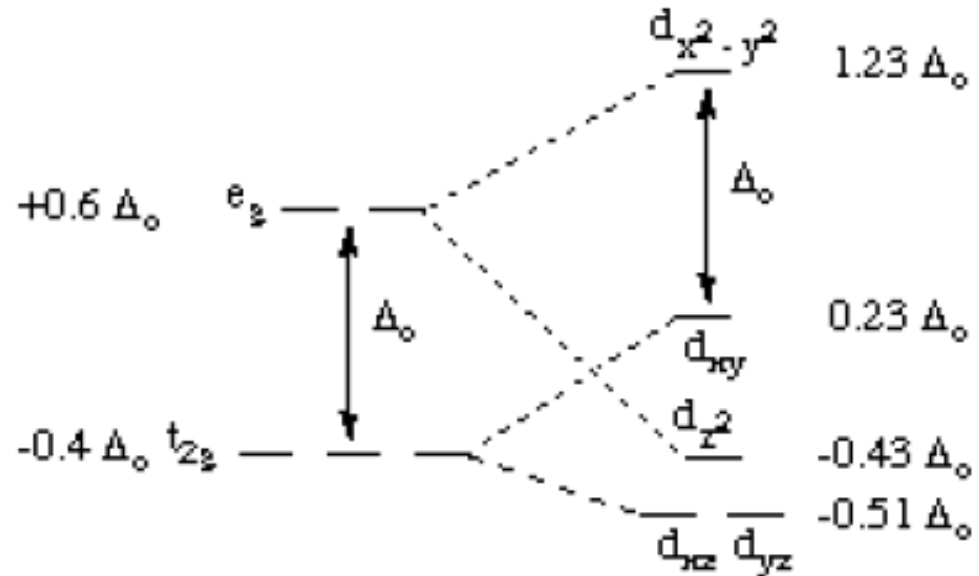


d⁸ square planar system



- 5) δ_1 is large in $[\text{Ni}(\text{CN})_4]^{2-}$ due to CN^- . **Effect of Ligand field is manifested here**
- 6) δ_1 is large in Pt^{2+} , Pd^{2+} (d⁸) systems due to size of metal ion. Most of the complexes of these metal ions are square planar in nature even with weak ligands. **manifestation of the effect of the metal atom/ ion**
- 7) Square planar complexes can thus be assumed to be the extreme case of tetragonal elongation of octahedron. Two of the octahedral ligands are at infinite distance from the center of the octahedron.

Square planar energy level



Octahedral
coordination

Square
planar
coordination
(xy plane)

d^8 system

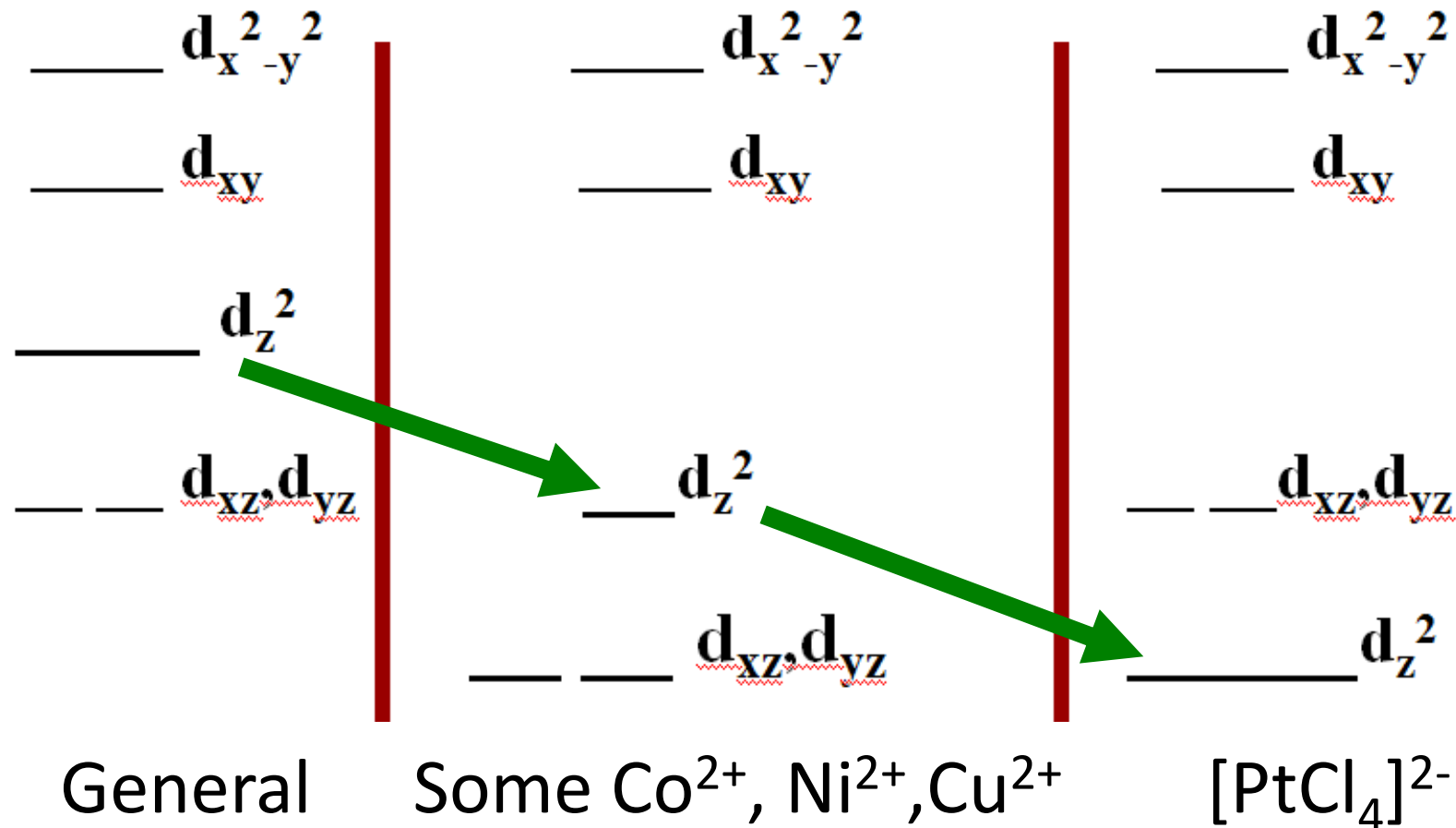
$$\begin{aligned} \text{CFSE}_{\text{sq. pl.}} &= 4 \text{ electrons } (-0.51 \Delta_o) + 2 \text{ electrons } (-0.43 \Delta_o) \\ &\quad + 2 \text{ electrons } (+0.23 \Delta_o) + P \\ &= -2.44 \Delta_o + P \end{aligned}$$

- The d_{z^2} orbital falls the most, as its electrons are concentrated in lobes along the z-axis.
- The d_{xz} and d_{yz} orbitals also drop in energy, but not as much.
- Conversely, the $d_{x^2-y^2}$ and the d_{xy} orbitals increase in energy.

Change in square planar energy level



If the tetragonal distortion (δ_1, δ_2) is large then d_z^2 orbital can lie very low.

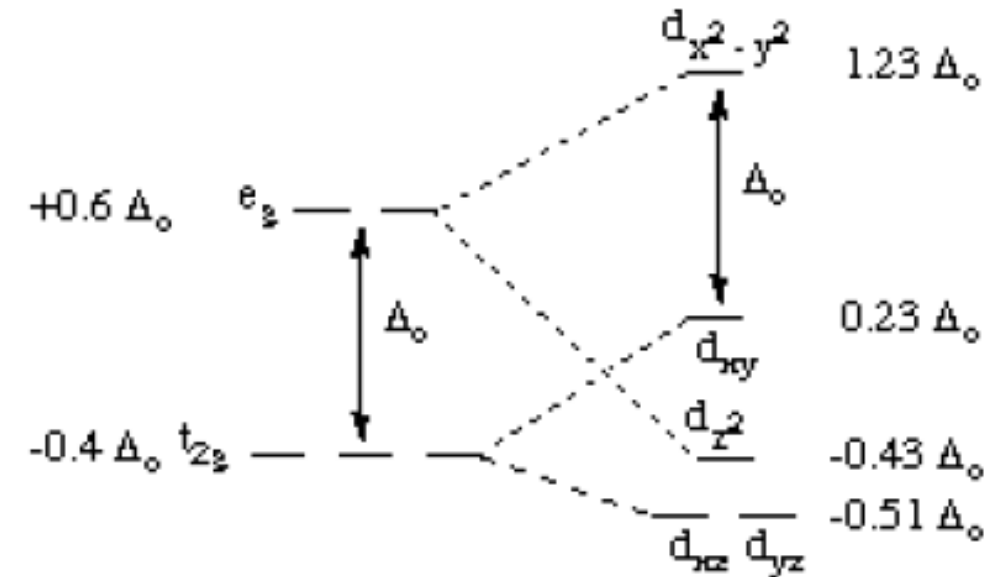


Preferred geometry of Cu(II) Complex



Cu^{2+} ion (d^9 system) has a much higher CFSE value in a square planar configuration ($\text{CFSE} = -1.22 \Delta_o$) than in octahedral ($-0.6 \Delta_o$) or tetrahedral configuration ($\text{CFSE} = -0.18 \Delta_o$)

CFSE values supports why Cu^{2+} ions forms **square planar complexes** rather than tetrahedral or octahedral complexes in both the fields



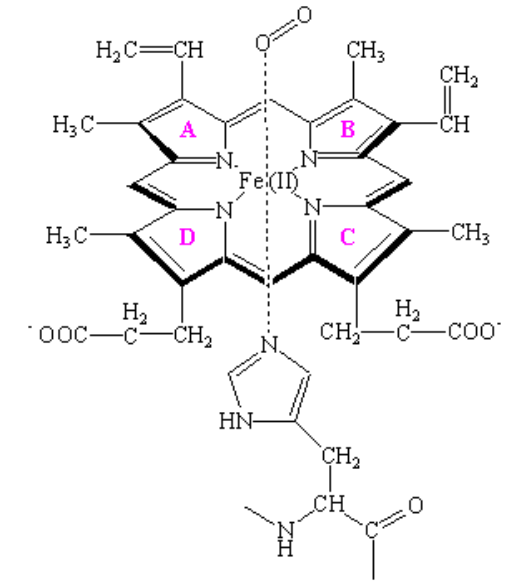
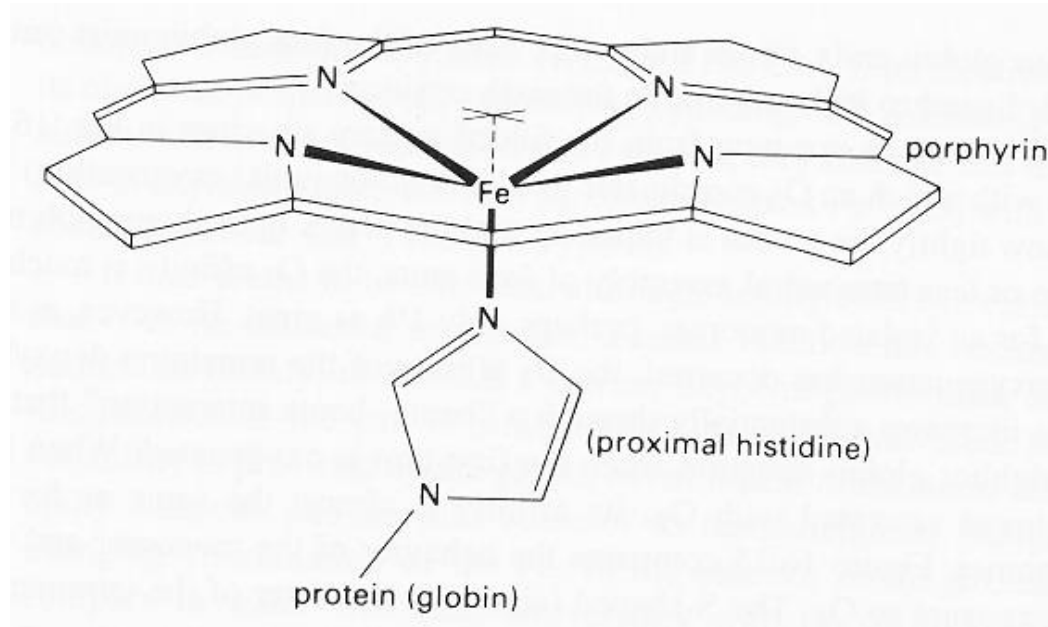
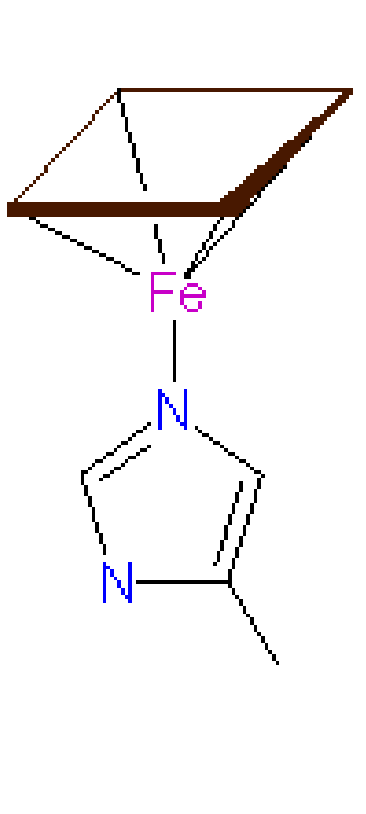
Square planar system



d ⁴	Cr ²⁺	4 unpaired electrons	(d _{yz} , d _{zx}) ² (d _{z2}) ¹ (d _{xy}) ¹
d ⁶	Fe ²⁺	2 unpaired electrons	(d _{yz} , d _{zx}) ⁴ (d _{z2}) ¹ (d _{xy}) ¹
d ⁷	Co ²⁺	1 unpaired electron	(d _{yz} , d _{zx}) ⁴ (d _{z2}) ² (d _{xy}) ¹
d ⁸	0 unpaired electrons		
	Ni ²⁺ , Rh ⁺ , Ir ⁺	Strong ligands	
	Au ³⁺ , Pt ²⁺ , Pd ²⁺	Strong or weak ligands	
d ⁹	Cu ²⁺	1 unpaired electron	(d _{yz} , d _{zx}) ⁴ (d _{xy}) ² (d _{z2}) ² (d _{x2-y2}) ¹

It shows the filling of electrons in the d-orbitals for the complexes shown above if these would have square planar geometry only.

Understanding the structure of Hemoglobin



Sample problems



Q: Identify the reason behind distortion in the complexes $[\text{MnF}_2\text{Cl}_4]^{4-}$ and $[\text{Mn}(\text{CN})_6]^{4-}$. (ii) Write their expected geometry (with elongation/ compression) and justify briefly for both cases.

Q1(i) Identify the reason behind distortion in the complexes $[\text{MnF}_2\text{Cl}_4]^{4-}$ and $[\text{Mn}(\text{CN})_6]^{4-}$. (ii) Write their expected geometry (with elongation/ compression) and justify briefly for both cases.

- (i) Natural distortion and Jahn-Teller distortion for $[\text{MnF}_2\text{Cl}_4]^{4-}$ and $[\text{Mn}(\text{CN})_6]^{4-}$ respectively.
- (ii) Tetragonal compression is expected for $[\text{MnF}_2\text{Cl}_4]^{4-}$ and tetragonal elongation is possible for $[\text{Mn}(\text{CN})_6]^{4-}$. $[\text{MnF}_2\text{Cl}_4]^{4-}$ has a natural distortion due to different kinds of bonding, (Mn-F and Mn-Cl). As Cl^- ligand is larger in size, Mn-Cl bonds will be elongated resulting in tetragonal compression. In $[\text{Mn}(\text{CN})_6]^{4-}$ Mn^{2+} ion is d^5 configuration and CN^- is a strong ligand field. So, $t_{2g}^5 e_g^0$, asymmetry in t_{2g} orbital mild Jahn-teller distortion. In tetragonal elongation $\text{CFSE} = -2/3\delta_2$ but in compression, $\text{CFSE} = -1/3\delta_2$

Example: d^8 system



$[\text{Ni}(\text{CN})_4]^{2-}$ - diamagnetic

Here Nickel is in +2 oxidation state;

$\text{Ni}^{+2} \rightarrow d^8$ and diamagnetic $\rightarrow$ Square Planar ;

Here this is entirely due to CN^- being a strong ligand.

High $\Delta_o \rightarrow$ high δ_1

$\text{Ni}(\text{CO})_4$

Example: d⁸ system



[Ni(CN)₄]²⁻ - diamagnetic

Here Nickel is in +2 oxidation state;

Ni⁺² → d⁸ and diamagnetic → Square Planar ;

Here this is entirely due to CN⁻ being a strong ligand.

High Δ_o → high δ_1

Ni(CO)₄ - diamagnetic

Here Ni⁰ → d¹⁰ (No 4s electrons)

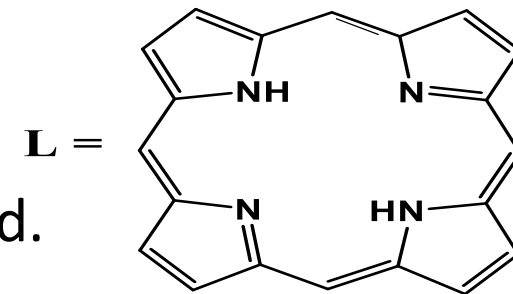
Therefore, it has to be diamagnetic as there is no unpaired electron in the metal orbitals

It has a tetrahedral geometry (**CANNOT BE EXPLAINED BY CFT**)

Sample problems



Q2 The general formula of a red-colored Ni(II) complex is $\text{NiL}(\text{ClO}_4)_2$, where L is a neutral macrocyclic planar ligand as shown next. When the thiocyanate (SCN^-) ions are added, the violet complex $\text{NiL}(\text{SCN})_2$ is formed. Answer the following questions in tabular form.

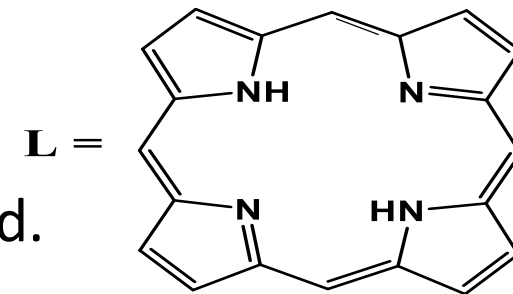


General formula of complex	Geometry of the complex	Spin-only magnetic moment, BM	Molar conductivity
$\text{NiL}(\text{ClO}_4)_2$			4 charges
$\text{NiL}(\text{SCN})_2$			0 charges

Sample problems



Q2 The general formula of a red-colored Ni(II) complex is $\text{NiL}(\text{ClO}_4)_2$, where L is a neutral macrocyclic planar ligand as shown next. When the thiocyanate (SCN^-) ions are added, the violet complex $\text{NiL}(\text{SCN})_2$ is formed. Answer the following questions in tabular form.



General formula of complex	Geometry of the complex	Spin-only magnetic moment, BM	Molar conductivity
$\text{NiL}(\text{ClO}_4)_2$	Square planar	0	4 charges
$\text{NiL}(\text{SCN})_2$	Octahedral	2.82	0 charges